

TactileNet: Bringing Touch Closer in the Digital World

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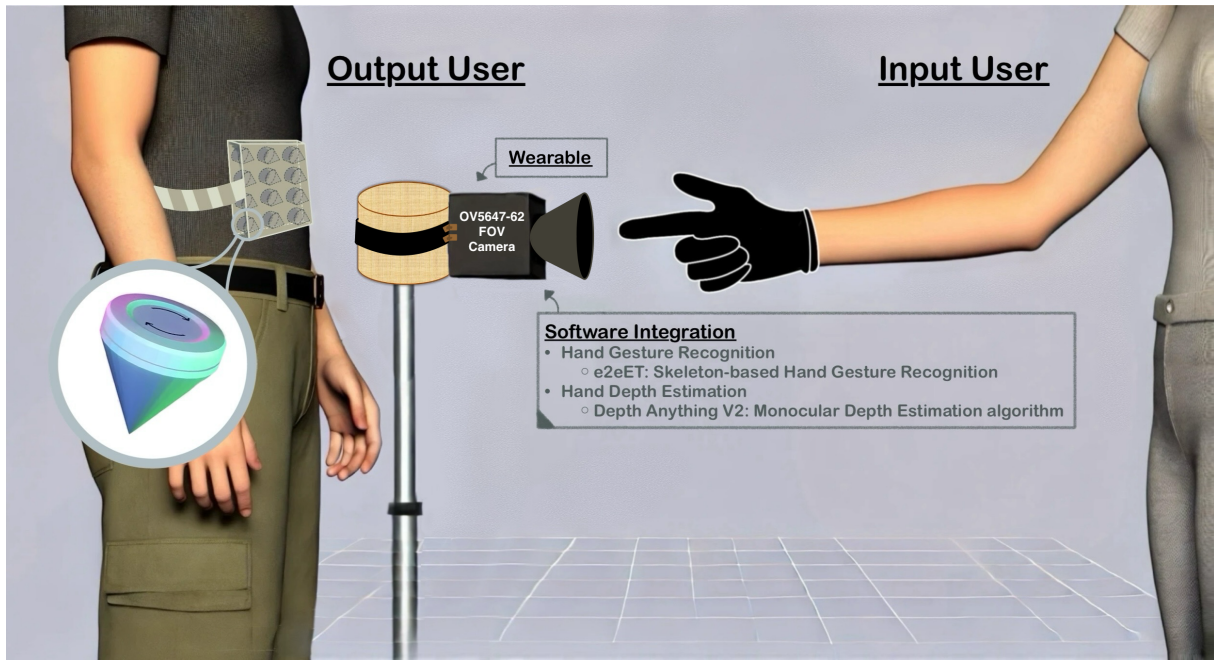


Figure 1: Visual representation of TactileNet system

ABSTRACT

Haptic technologies have improved human-computer interaction (HCI) by enabling immersive tactile experiences. This proposal introduces a 2D lateral touch actuator to simulate rubbing sensations on human skin and transmit them over the internet. Using a controllable tilted roller mechanism, the system recreates various touch sensations. Advanced hand gesture recognition algorithms

identify and transmit the input user's actions to the output user. Utilizing AI-enabled computing chips from the Gen-M kit, the project aims to develop a prototype that accurately simulates natural touch interactions.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

*Both authors contributed equally to this research.

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1 INTRODUCTION

In recent years, the advent of advanced haptic technologies has significantly enhanced the field of human-computer interaction (HCI). These technologies have expanded the possibilities for creating immersive and tactile user experiences, fostering a deeper connection between users and digital environments. For example, Wu and Culbertson’s work on wearable haptic devices demonstrates the potential of such technologies in providing realistic touch feedback, which is crucial for social and emotional communication in virtual settings [2]. Furthermore, Jalapati et al. highlighted the importance of crossmodal systems in affective communication, showing how vocal expressions can be effectively transposed into haptic signals to enhance emotional connections between users [1].

As illustrated in Figure 1, the actuator of the proposed system employs a controllable tilted moving roller mechanism that mimics the sensation of a lateral rubbing touch. We plan to map the gesture as the direction of the touch and the depth of the hand as the intensity, which can be achieved by varying the angular velocity of the rollers. More specifically, for the intensity, closer to the camera means higher strength, and vice versa.

By manipulating the rollers’ tilt and movement, the actuator can recreate various touch sensations on the user’s skin. The touch sensation simulation input is provided remotely through two advanced algorithms: hand gesture recognition and hand depth estimation. The hand gesture recognition algorithm determines the touch direction initiated by the input user, adjusting the actuator’s roller direction accordingly. Simultaneously, the depth estimation algorithm modulates the touch intensity by varying the roller’s angular speed. Both directional and intensity adjustments are then transmitted to the receiving actuator for the output user. The primary objective of this project is to develop a robust prototype that accurately simulates natural touch interactions, thereby enhancing remote sensory communication and interaction capabilities.

This project uses AI-enabled computing chips provided in the Gen-M kit to implement hand gesture recognition and hand depth estimation algorithms. The system’s potential applications are vast, ranging from enhancing remote personal interactions to providing therapeutic touch experiences in telemedicine.

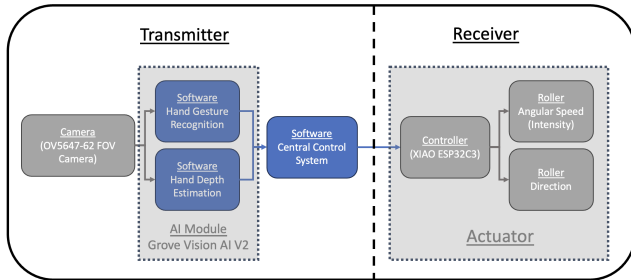


Figure 2: Flow-chart showing sub-components in the system

2 HARDWARE DESIGN & SOFTWARE IMPLEMENTATION

The implementation of our 2D lateral touch actuator involves several components to ensure realistic touch simulation. The core of the

system is the controllable tilted moving roller mechanism, which is designed to replicate the sensation of sweeping touch on human skin. Each actuator unit consists of multiple controllable tilted rollers that can be adjusted together to create various sweeping direction. The actuators are arranged in a grid pattern. The system can also includes sensors to detect the direction and intensity of the touch, providing feedback for real-time adjustments.

2.1 Hardware Integration

For the transmitter side, as illustrated in Figure 1, the system employs the Grove Vision AI Module for both AI algorithms, allowing for accurate interpretation of the touch direction and intensity from the input user. On the receiver side, the system’s controller is powered by the Seeed Studio XIAO series, which manages the servo motors and sensors. It translates the received signals into corresponding touch sensations and intensities by controlling the angular speed and direction of the roller on the output user’s skin. These ESP32 family microcontrollers are programmed using Arduino to ensure precise control over the actuators.

2.2 Software Integration

The software integration for this project involves the seamless combination of two primary algorithms: hand gesture recognition and hand depth estimation. By utilizing the state-of-art machine learning technique, these algorithms will work together to interpret the hand gestures and their proximity to the camera, translating these inputs into haptic feedback for the user. The mapping consists of two parts: the gesture corresponds to the sweeping direction on the skin, and the distance between the hand and the camera is mapped to the intensity of the rubbing. Precise coordination between the hardware and software sub-components is expected. It is critical for achieving a realistic and immersive sensory experience.

2.2.1 Hand Gesture Recognition: Hand gesture recognition is the first part for recording the input of touch simulation. The algorithm that we choose to use is the *e2eET*: Skeleton-based Hand Gesture Recognition using Data-Level Fusion algorithm. This state-of-the-art model, recognized for its high accuracy (95.83%) on the DHG-14 Benchmark, is ideal for our application[4]. The process begins with real-time capture of hand movements using a camera. These movements are then converted into a skeletal representation, suitable for further processing like gesture classification and tracking by the *e2eET* model. Once the gestures are classified, the number of extended fingers and detected hand patterns will be used to define the type and area of the touch simulation. Besides, by tracking the hand movement in the camera frame and mapping it to the actuator frame, this algorithm can provide the direction and speed input to the touch actuator. An API will be developed to transmit the recognized gestures to the central control system for further processing, ensuring accurate and reliable data flow.

2.2.2 Hand Depth Estimation: The second part for the touch simulation input is the hand depth estimation. It will be handled by the *Depth Anything V2*: Monocular Depth Estimation algorithm. This model is chosen for its precision in estimating the distance of the hand from the camera [3], which will be mapped to the force of the touch. Real-time video frames from the camera will be processed to

estimate the hand's depth in each frame. A function will then map the estimated depth to a corresponding force value, with the closer the hand is to the camera, the greater the force applied. To facilitate this, an API will be developed to transmit in-depth information to the central control system, allowing for seamless integration with the gesture recognition data. It's worth noting that as our project focuses on hand depth estimation, fine-tuning and training will be considered during implementation.

2.2.3 Software Central Control System. The central control system is critical for integrating data from the hand gesture recognition and depth estimation algorithms. It will collect real-time data streams from both sources, ensuring synchronized data processing. The system will then fuse this data to generate comprehensive touch commands. A data fusion algorithm will be developed to integrate outputs from the *e2eET* and *Depth Anything V2* models, translating these into precise control signals for the haptic actuators.

2.3 Prototype Development

The initial prototype will be developed using the *Gen-M kit* provided by *Seeed Studio*. This kit includes all necessary components, such as microcontrollers, camera sensors, and rotary actuators, to build the touch simulator.

The development process will begin with hardware setup, assembling the camera and haptic actuators as per design specifications and ensuring reliable connectivity between all components. To support the controlled tilting movement of the motor array, a pair of

servo motor will be implemented to control the degree of freedoms in x and y directions. For simplicity, the rollers are not allowed to tilt independently, all movements are synced.

This software implementation integrating the *e2eET* and *Depth Anything V2* models into the provided AI enabled computation chip. Extensive testing with users will be conducted to refine the touch simulation, incorporating user feedback to enhance the realism and effectiveness of the experience. Both the *Depth-Anything-V2-Small* and the *e2eET* models will run on the *Grove Vision AI V2* module.

The user experience study will be a critical part of the prototype development. User testing sessions will be conducted to evaluate the realism and effectiveness of the simulation. Adjustments will be applied to both hardware and software components based on user input to enhance the simulation fineness and ensure a more immersive experience.

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